

Supanart (Mine) Barnsongkit

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EDUCATION

Columbia University

Master of Science in Computer Science, Machine Learning track

New York, NY
Expected Dec 2027

Chulalongkorn University

Bachelor of Engineering in Computer Engineering, GPA: 3.73/4.00, First Class Honours

Bangkok, Thailand
Aug 2019 – May 2023

- No. 1 University in Thailand by QS World University Rankings 2026
- Thesis: End-to-End Machine Learning Lifecycle Management Platform
- Relevant Coursework: Computer Vision, Pattern Recognition, Data Structures and Algorithms

EXPERIENCE

Senior Software Engineer

Arise by INFINITAS (subsidiary of Krungthai Bank, 2nd largest bank in Thailand by assets)
CLICX Bank - Thailand's first virtual bank (Neobank)

May 2024 – Aug 2026
Bangkok, Thailand

- Led a 5-person SWE team in designing and implementing backend services and APIs for the user-onboarding and bank-account-opening journeys of the CLICX mobile app using Go, Kafka, Redis, MongoDB, and PostgreSQL
- Owned the end-to-end lifecycle of onboarding APIs, including deployment to Kubernetes with ArgoCD, go-live, production monitoring with Prometheus and Grafana, and incident investigation
- Reduced response latency across onboarding APIs from approximately 2 seconds to 20 milliseconds by improving caching and optimizing MongoDB indexes for per-user documents

Software Engineer Team Lead

Protomate (AI-enabled motorcycle helmets to improve rider safety)

Jan 2023 – Feb 2024
Bangkok, Thailand

- Managed the entire software engineering team (7-person) responsible for implementing the integrated software platform for Thailand's first smart helmet, its Flutter mobile app, and nRF-based remote control
- Owned the software roadmap and aligned roadmap priorities with mechanical and electrical engineering leads
- Designed device-to-app connectivity using Bluetooth Classic and BLE, implemented Secure Boot, and integrated the mobile app and remote control with custom Android firmware

Machine Learning Engineer Intern

OxygenAI (AI company building computer-vision products for retail and smart-city)

June 2022 – July 2022
Bangkok, Thailand

- Researched object detection in fisheye images without prior dewarping, addressing challenges from lens distortion and varying object orientations
- Developed a computer vision model by modifying YOLOv5 to correct bounding-box orientation using rotation-aware regression

PROJECTS

Rainfall Nowcasting on Thai Department of Drainage and Sewerage Radar Data

May 2022

Data Science and Engineering Project

- Created a real-time rainfall nowcasting system for Bangkok with an end-to-end data pipeline and dashboard using GCP (Cloud Functions, Pub/Sub, Cloud Storage), TensorFlow, Folium, and Power BI
- Led the team to train and develop optical flow, ConvLSTM, and RainNet (autoencoder) models using TensorFlow and Keras, achieving MSE of less than 0.001 on radar- and wind-data scraped from DDS data

Vehicle Detection and Counting

Jul 2021

Individual Research Study on Machine Learning

- Built a Mask R-CNN system to detect and segment vehicles and estimate traffic volume from CCTV data using PyTorch and OpenCV, achieving 60.2% box AP50 and 56.6% mask AP50

AWARDS & HONORS

RoboCup@Home 2022, Open Platform League — World 1st Runner-up

- Built "Walkie," a household service robot. 11 teams from 7 countries qualified for the final
- Implemented pre-labeling models for the open-vocabulary object detection tasks using ViLD method (Vision and Language Knowledge Distillation). Assisted the sub-team in implementing person-tracking and pose-estimation algorithms

TECHNICAL SKILLS

ML Libraries: PyTorch, TensorFlow, NumPy, pandas, Matplotlib, OpenCV

Languages: Python, Go, Kotlin, Java, C/C++, SQL (Postgres), TypeScript, JavaScript

Frameworks: FastAPI, Gin, Fiber, Spring, React, Next.js, Node.js

DevOps & Cloud: Git, Docker, Kubernetes, Argo CD, Prometheus, Grafana, AWS, GCP